

# Electrical Topology Diagram (Implementation v6)

As-of date: 2026-07-19

Purpose: provide a complete, install-level electrical topology for the current build scope, including all major electrical components, fuse IDs, fuse housings, planned wire gauges, and estimated one-way run lengths for procurement planning.

Related docs:

- Canonical electrical/system baseline: docs/core/SYSTEMS.md
- Detailed fuse matrix: docs/implementation/ELECTRICAL\_fuse\_schedule.md
- Battery and trunk recalculation record: docs/studies/ELECTRICAL\_battery\_fuse\_wire\_recalc\_2026-02-18.md
- Decisions and unresolved items: docs/core/TRACKING.md
- Procurement source of truth: bom/bom\_estimated\_items.csv

## Sweep Outcomes Included In This Revision

- Keeps alternator charging architecture on the dedicated 48V secondary alternator path ( Mechman + WS500 + APM-48 baseline); obsolete pre-Mechman charger paths are removed from primary topology.
- Keeps Lynx Slot 3 branch to alternator input with F-04 150A ( 58V/80V MEGA).
- Removes obsolete pre-Mechman engine-bay fuse/conductor placeholders from active architecture.
- Clarifies WS500 fused-lead visibility ( F-12/F-13 ) and treats current-sense high/low as an unfused twisted sense pair per current Wakespeed manual.
- Adds Ford Upfitter #3 -> F-15 -> WS500 brown ignition manual alternator-control path.
- Adds the detailed Mechman/WS500/APM-48 install guide as the shop reference for staged installation, first-run checks, and load-dump/shutdown handling.
- Defines APM-48 as a parallel surge clamp at the alternator rather than a series charge-current device.
- Added explicit fuse-holder/housing definitions for every fuse family ( Class T , Lynx MEGA , inline MIDI/ANL/AMI , PV gPV , and ATO/ATC ).
- Added conductor schedule across 48V , 12V , PV, and AC segments with explicit assumptions.
- Updated 12V topology to a shared 12V junction fed by an Orion-Tr Smart 48/12-30 charger and a 12V 100Ah buffer battery branch, with F-11 source fuse plus SW-12V-BATT manual isolation.
- Added a full-circuit estimated run-length validation pass ( C-01 through C-40 ) and purchase-ready wire rollup totals.

## Current Commissioning Snapshot ( 2026-05-27 )

- 48V bus live-tested at 55.5V throughout the system, including at the MultiPlus.
- MultiPlus-II inverter mode tested with inverter light on, slight normal hum, and no reported errors.
- SmartShunt and Orion-Tr Smart connected in VictronConnect.
- Cerbo GX access point/remote-console workflow active; Cerbo is powered from a small inline fused 48V feed and connected to MultiPlus via VE.Bus RJ45.
- Short AC-in shore-charge test passed at household-source current limits: about 1294W shore input and about 54.3V x 21.6A battery charging in bulk.
- Hold open: MultiPlus LiFePO4 charge profile is now programmed/owner-verified by supervised first-battery behavior; AC-out branch/GFCI, alternator commissioning, Cerbo hard-mounting, and final strain-relief/abrasion-control remain future gates.

## Current Physical Installation Snapshot ( 2026-07-19 )

- The electrical module is hard-mounted through the finished Lonseal/plywood floor into registered truck-bed hardpoints. It fits cleanly and is sufficiently stable for continued stationary installation; the planned Bench tie-in remains required for final anti-rack stiffness and road restraint.
- The MultiPlus and combined AC breaker enclosure were removed before lifting to reduce module weight. Embedded pronged/spiked T-nuts in the plywood backer allow direct front-side remounting without loose nuts behind the board.
- All three batteries' positive and negative 2/0 AWG branch cables are cut, lugged, heat-shrunk, and landed at the battery-side busbars. The batteries remain electrically isolated from one another until Batteries 2 and 3 are individually charged, rested, and matched within 0.1V before paralleling.
- The 12V buffer-battery branch is near physical completion; keep its negative direct to the fuse-panel main negative stud, not through SW-12V-BATT . The positive path remains F-11 -> SW-12V-BATT -> panel main + .

### Orion fuse discriminator — do not combine these positions

- F-06 : **one standalone inline fuse on Orion 48V input positive**, final 20A/80V Littelfuse FKS/ATO fuse and matching holder; existing 30A/58V MIDI is interim only.

- Lynx Slot 2 / F-03 : **MPPT branch**, 60A/80V MEGA. If the physically X-marked/misrated Lynx fuse is in Slot 2, replace it with the purchased Victron 60A/80V MEGA; it is not the Orion input fuse.
- Lynx Slot 4 / F-05 : **open spare**. If the X-marked position is Slot 4, remove/leave it empty; it is not a second Orion input fuse.
- F-07 : **Orion 12V output positive**, 60A/80V MEGA in the separate Victron holder near the Orion.
- F-11 : **12V buffer-battery positive**, 100A ANL near the battery, upstream of SW-12V-BATT .
- No 32V -rated fuse is acceptable on any energized 48V branch. Identify the physical X-marked fuse by Lynx slot before replacing or discarding it.

## Length Estimation Defaults Used In This Pass

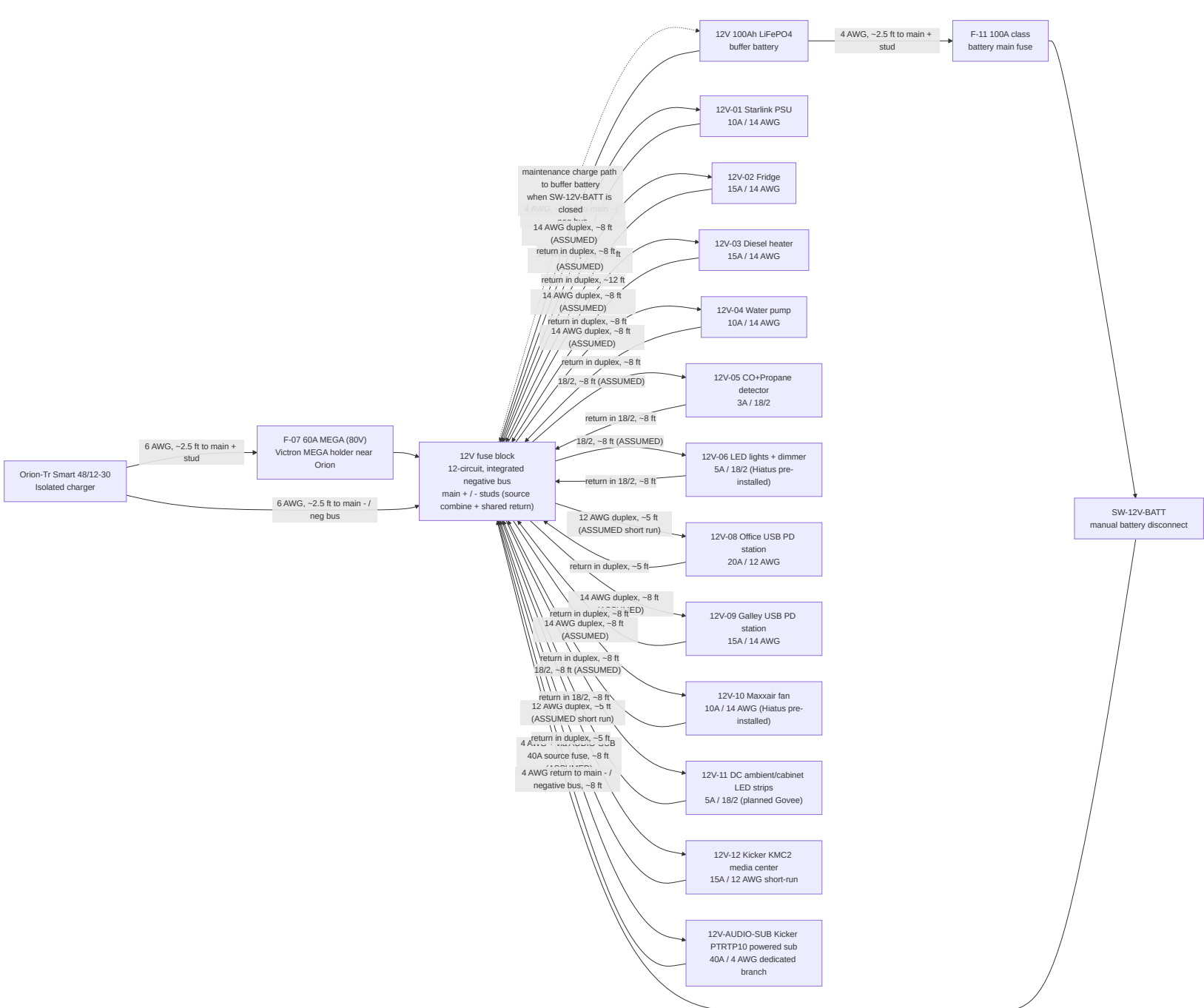
1. Cabinet internal interconnect default: 2.5 ft one-way ( ASSUMED ).
2. Cabinet-to-near load branch default: 8 ft one-way ( ASSUMED ).
3. Cabinet-to-far load branch default: 12 ft one-way ( ASSUMED ).
4. AC branch to receptacle chain default: 15 ft one-way per branch leg ( ASSUMED ).
5. Policy lock: use the smallest gauge that meets current and voltage-drop targets; do not auto-upsize, but flag warnings when margin is tight.
6. Parallel battery bank lock: keep each battery's **total positive + negative path resistance** similar. Equal positive-only length is not required if the positive-length differences are offset by negative-length differences; do not add unnecessary 2/0 cable coils solely for cosmetic equality.

## Battery Fuse/Wire Recalculation Basis (2026-02-18 + 2026-03-19 sync)

- Scope in this pass is limited to battery-side and major 48V trunk paths ( C-01 through C-15 ).
- Provisional battery listing inputs used: 51.2V 100Ah ,  $\leq 200A$  current limit per battery.
- Conservative sizing factors used in this pass:
  1. Parallel-sharing factor  $K_{share} = 1.5$
  2. Continuous margin factor  $K_{cont} = 1.25$
- Current/interim envelope used for battery-discharge branch sizing in current architecture:  $I_{total} = F-02 + F-06 = 125A + 30A = 155A$  ; final cleanup envelope after the Orion downsize is  $125A + 20A = 145A$  .
- Per-battery design current: interim  $I_{batt\_design} = (155A / 3) * 1.5 = 77.5A$  ; final  $I_{batt\_design} = (145A / 3) * 1.5 = 72.5A$  .
- Continuous-adjusted minimum battery branch fuse threshold: interim  $77.5A * 1.25 = 96.9A$  ; final  $72.5A * 1.25 = 90.6A$  .
- Provisional battery branch fuse selection: F-01A/B/C = 200A Class T , constrained by the provisional battery  $\leq 200A$  current-limit listing.
- Final lock gate: validate true 51.2V battery datasheet/manual current and terminal limits before permanent fuse lock; if lower limits are confirmed, move to 175A .
- Cable procurement remains estimate-based until CAD/field run lengths are frozen. This pass sets a no-padding 2/0 estimate baseline of 77.5 ft total ( 42.5 ft red, 35.0 ft black), including the dedicated alternator migration path.

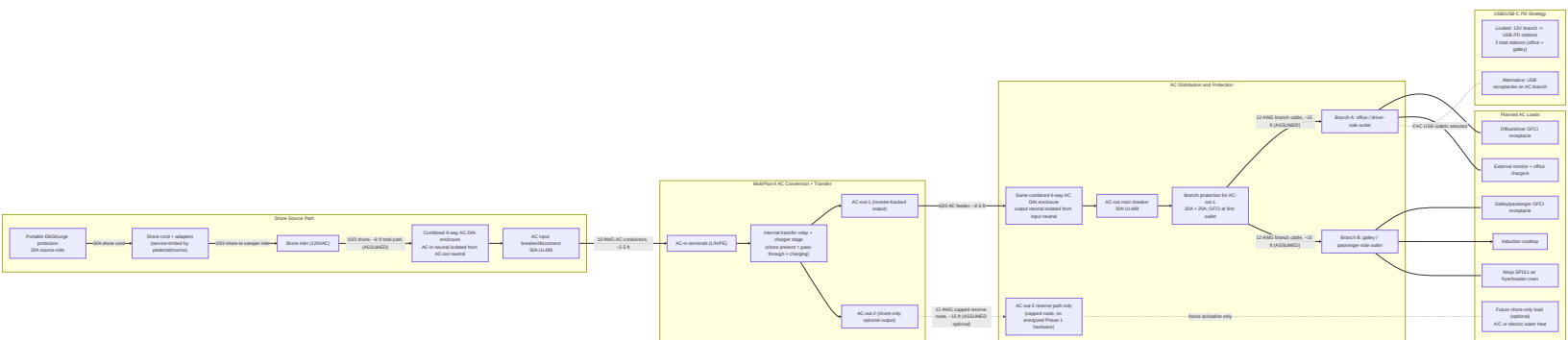
## Complete Power Topology (48V Core + Charge Sources)





## 12V Operating Intent (Locked)

- Orion-Tr Smart 48/12-30 is the primary 12V charger/feed source.
- SW-12V-BATT is **normally closed** in operation; open is service/isolation mode only.
- With SW-12V-BATT closed, Orion output at the fuse-block main + stud maintains/charges the 12V buffer battery through the shared-junction path.
- With SW-12V-BATT open, the buffer battery is isolated from the main + stud; this mode is for service/troubleshooting only and not the default operating state.
- The buffer battery remains in the active operating path during normal use and is intended to absorb transients/peaks on the 12V rail.
- The fuse block is the 12V junction device in this baseline: main + stud is the source-combine point, and the integrated negative bus/main - is the shared return point.
- Do not solder-splice high-current source conductors; terminate with crimped lugs on rated studs/junction hardware.
- Hiatus pre-installed 12V branches are tracked here as 12V-06 (factory LED lights with dimmer) and 12V-10 (Maxxair fan); verify final installed branch labeling during electrical audit.
- Planned Govee/ambient strip lighting is a separate branch ( 12V-11 ) and not part of the Hiatus factory 12V-06 lighting circuit.
- DC-first lighting intent: keep ambient/cabinet strip lighting on 12V-11 so nighttime lighting does not require inverter operation.
- Camper audio intent: Kicker 46KMC2 source unit uses a 15A 12V-12 branch from the fuse block, while Kicker 49PTRTP10 powered sub uses a separate 40A fused 4 AWG branch from the 12V source/junction rather than a normal blade-fuse panel branch. Bass peaks may exceed the Orion 30A continuous feed and are supported by the 12V buffer battery for short durations.



### AC Operating Behavior (Expected)

- Shore present: MultiPlus transfer relay closes, AC-in is passed to AC-out paths, and charger stage charges the 48V bank.
- Shore absent: MultiPlus transfers to inverter mode and powers AC-out-1 from battery; AC-out-2 drops by design.
- AC-in hardware is 30A ( source/adapters -> portable EMS -> shore cord -> L5-30 inlet -> 30A breaker -> 10 AWG AC-in conductors ); set MultiPlus input current limit to actual source ( 15A , 20A , or 30A ) to avoid pedestal/source breaker trips.
- Initial battery charging may use AC-in-only mode with AC-out branch loads disconnected.

## AC Safety/Protection Chain (What Must Exist)

- Upstream shore protection chain before MultiPlus AC-in: shore source/adapters -> portable EMS -> shore cord -> shore inlet -> 30A AC input breaker/disconnect.
- AC-out protection chain: MultiPlus AC-out-1 -> 10/3 feeder -> 30A AC-out main breaker -> 20A branch breakers -> GFCI receptacles.
- Single-enclosure DIN architecture with isolated AC-in and AC-out neutral paths and no neutral mixing.
- Continuous equipment grounding path from shore inlet through MultiPlus and branch circuits, plus chassis bond in mobile install context.
- Neutral/ground handling must follow MultiPlus relay behavior; do not add an always-bonded downstream neutral-ground bond in branch receptacle wiring.

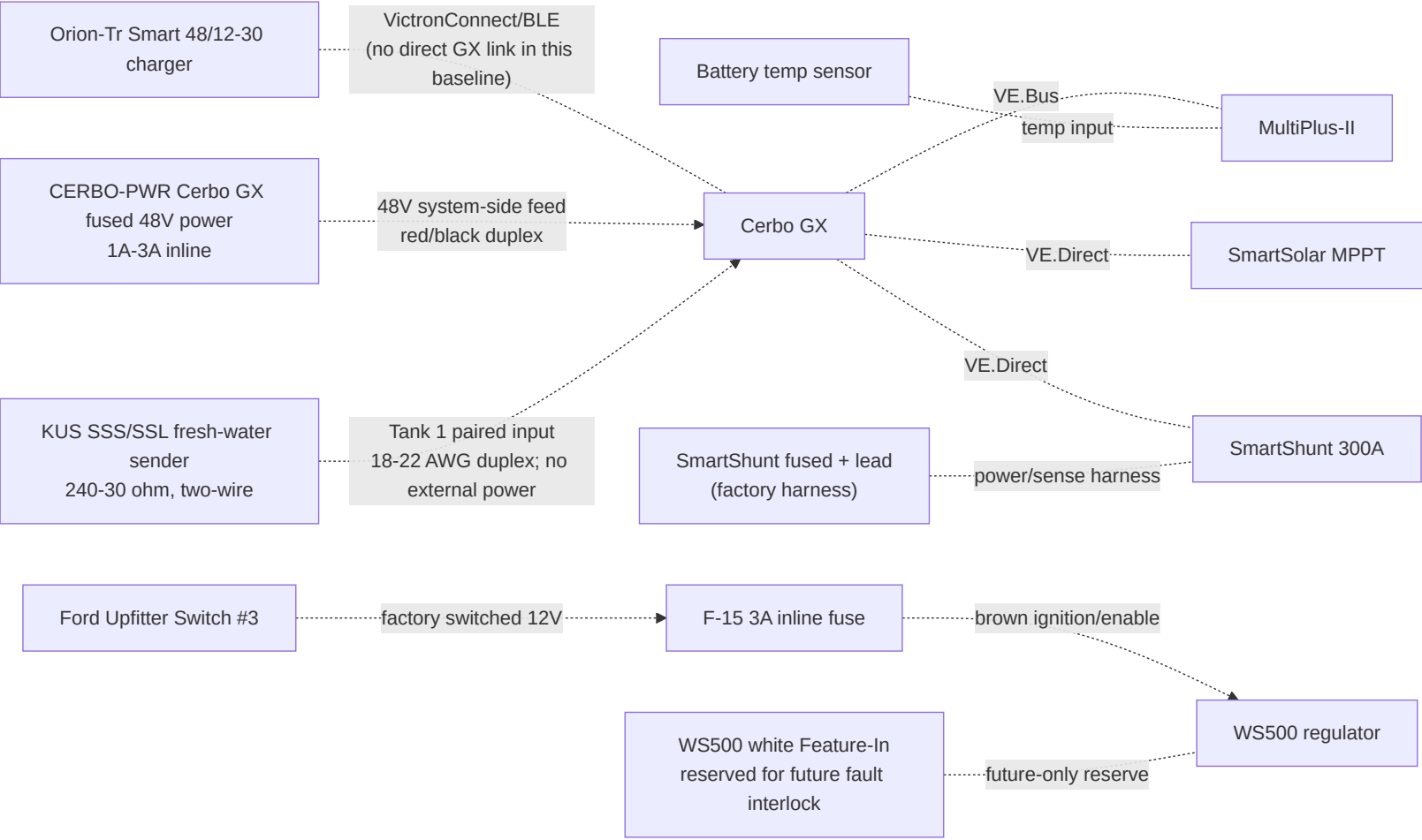
### AC Reference Basis (Manufacturer Guidance)

- Victron MultiPlus-II 120V installation guidance ( AC-in breaker sizing, UL943-class residual-current protection on outputs, and AC-out-2 shore-only behavior): [https://www.victronenergy.com/media/pg/MultiPlus-II\\_120V/en/installation.html](https://www.victronenergy.com/media/pg/MultiPlus-II_120V/en/installation.html)
- Victron MultiPlus-II datasheet ( 48/3000/35-50 baseline model reference): <https://www.victronenergy.com/upload/documents/Datasheet-MultiPlus-II-inverter-charger-120V-EN.pdf>

## AC/USB Baseline Locked For BOM

- Shore interface: 30A RV-style inlet baseline with adapter kit for 15A / 20A hookups.
- AC input protection: portable EMS + combined DIN enclosure + 30A AC input breaker/disconnect upstream of MultiPlus AC-in.
- AC-out-1 distribution: 30A AC-out main plus two protected 20A branches with GFCI-at-first-outlet strategy.
- Receptacle plan: current purchased baseline is 2 total 120V GFCI receptacles, one per active branch: Branch A = office/driver side, Branch B = galley/passenger side. Prior downstream non-GFCI receptacles are obsolete unless a later layout revision reopens them.
- USB charging plan: 2 DC-fed USB PD station assemblies on 12V branches ( 1 office, 1 galley) with branch baselines of 20A (office) and 15A (galley).
- AC-out-2 remains reserve-only in Phase 1 (labeled capped route; no energized branch hardware procured).

## Monitoring and Control Topology



## Fuse and Switch Housing Map (Where Each Item Is Physically Housed)

| Item ID   | Item value/type                                                   | Housing method                                                                                                                                                                    | Location                                                                                     |
|-----------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| F-01A     | 200A Class T (provisional)                                        | Blue Sea Class T fuse block (110A-200A family)                                                                                                                                    | Battery compartment near Battery A +                                                         |
| F-01B     | 200A Class T (provisional)                                        | Blue Sea Class T fuse block (110A-200A family)                                                                                                                                    | Battery compartment near Battery B +                                                         |
| F-01C     | 200A Class T (provisional)                                        | Blue Sea Class T fuse block (110A-200A family)                                                                                                                                    | Battery compartment near Battery C +                                                         |
| F-02      | 125A MEGA                                                         | Lynx integrated slot holder                                                                                                                                                       | Lynx Slot 1                                                                                  |
| F-03      | 60A MEGA (80V Victron replacement stock)                          | Lynx integrated slot holder                                                                                                                                                       | Lynx Slot 2                                                                                  |
| F-04      | 150A MEGA                                                         | Lynx integrated slot holder                                                                                                                                                       | Lynx Slot 3 (dedicated alternator branch)                                                    |
| F-05      | Not installed / open spare position                               | Lynx integrated slot holder left blank                                                                                                                                            | Lynx Slot 4; reserve for future branch, not Orion                                            |
| F-06      | Orion 48v input fuse: interim 30A 58V MIDI; final 20A 80V FKS/ATO | Current build uses existing MIDI holder/fuse from a Lynx 48v+ bus tap; final cleanup stock purchased 2026-06-01: 3x Mouser 576-166.7000.5202 fuses + 3x 576-178.6150.0001 holders | Electrical cabinet between Lynx bus tap and Orion 48v +; keep source-side unfused lead short |
| F-07      | 60A MEGA (80V Victron replacement stock)                          | Victron MEGA fuse holder                                                                                                                                                          | Electrical cabinet at Orion 12v + source end                                                 |
| F-09A/B/C | 15A gPV each                                                      | 10x38 touch-safe fuse holders in PV combiner                                                                                                                                      | Roof-entry combiner enclosure                                                                |
| F-10      | Per branch (ATO/ATC)                                              | Integrated blade sockets in generic 12V fuse block                                                                                                                                | Electrical cabinet                                                                           |

|                          |                                                                        |                                                                                                            |                                                                                                                                                           |
|--------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| AUDIO-HU / 12V-12        | 15A source/head-unit branch; KMC2 harness also contains a 15A ATM fuse | 12V fuse block branch plus KMC2 harness fuse                                                               | Electrical cabinet to driver-side DC shelf/source face                                                                                                    |
| AUDIO-SUB                | 40A external fuse for Kicker PTRTP10 powered sub branch                | Inline/MRBF/AFS/ANL-class holder matched to selected 4 AWG kit; use 40A, not a generic 100A kit fuse       | Within about 18 in of the 12V source takeoff feeding the powered sub                                                                                      |
| F-11                     | 100A class (12V buffer battery main)                                   | Sealed inline MIDI/AMI/ANL holder                                                                          | Within ~`7"`` of 12V buffer battery positive post                                                                                                         |
| F-12                     | 10A/15A WS500 regulator power fuse                                     | Sealed inline holder; voltage rating must cover actual source voltage                                      | Near WS500 power lead source                                                                                                                              |
| F-13                     | 3A WS500 positive voltage-sense fuse                                   | Fuse/holder must be rated for actual 48v bank max voltage unless the WS500 harness rating proves otherwise | Near WS500 positive-sense source                                                                                                                          |
| CERBO-PWR                | 1A-3A Cerbo GX power fuse                                              | Small inline holder rated for the 48v bank maximum                                                         | Electrical cabinet near 48v system-positive takeoff; system side of disconnect preferred for bench shutdown                                               |
| WS500 current-sense pair | No fuse; purple/grey high/low sense pair to shunt/current-sense point  | Twist pair if extended; route away from noise                                                              | Shunt/current-sense source point                                                                                                                          |
| F-15                     | 3A WS500 ignition/enable control fuse                                  | Sealed inline ATC/ATO holder; 12V control circuit                                                          | Near Ford upfitter blunt-cut wire / WS500 control-wire handoff                                                                                            |
| SW-12V-BATT              | Manual battery disconnect switch                                       | Sealed rotary DC switch body                                                                               | Electrical cabinet near 12V fuse-block main + stud for service access                                                                                     |
| OEM-SHUNT                | External Victron-supplied inline fuse in red SmartShunt Vbatt+ cable   | Inline holder in supplied red cable                                                                        | Prefer battery-side positive if SOC continuity is desired with the main disconnect open; system side is acceptable for zero disconnect-off parasitic draw |

Retired from active architecture:

- Obsolete pre-Mechman charger/fuse paths are removed from active board layout and labels.

## Conductor Schedule (Start-to-Finish)

| Segment ID | Circuit segment      | Nominal voltage | Current basis                | Overcurrent protection | Planned wire gauge | Estimated one-way length (this pass)                 |
|------------|----------------------|-----------------|------------------------------|------------------------|--------------------|------------------------------------------------------|
| C-01       | Battery A + -> F-01A | 48V             | Battery branch, fuse-limited | F-01A 200A provisional | 2/0 AWG            | 2.5 ft planning placeholder; balance total loop path |
| C-02       | Battery B + -> F-01B | 48V             | Battery branch, fuse-limited | F-01B 200A provisional | 2/0 AWG            | 2.5 ft planning placeholder; balance total loop path |

|       |                                                                           |     |                                                                                                                                             |                                                       |                                              |                                                      |
|-------|---------------------------------------------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|----------------------------------------------|------------------------------------------------------|
| C-02C | Battery C + -> F-01C                                                      | 48V | Battery branch, fuse-limited                                                                                                                | F-01C 200A provisional                                | 2/0 AWG                                      | 2.5 ft planning placeholder; balance total loop path |
| C-03  | Class T outputs -> battery-side 48V + busbar -> disconnect input          | 48V | Combined trunk current                                                                                                                      | F-01A/B/C                                             | 2/0 AWG each branch                          | 2.5 ft each branch (ASSUMED, 4 conductors in rollup) |
| C-04  | Disconnect output -> Lynx + bus                                           | 48V | Aggregate discharge design current (155A interim = F-02 125A + Orion F-06 30A; 145A final after F-06 downsize to 20A)                       | Upstream Class T fuses                                | 2/0 AWG                                      | 2.5 ft (ASSUMED)                                     |
| C-05  | Battery negatives -> battery-side 48V - busbar -> SmartShunt battery side | 48V | Mixed-path rollup: 3x battery-negative branches at 77.5A interim design each + NEGBUS_TO_SHUNT trunk at 155A interim / 145A final aggregate | N/A (main negative path)                              | 2/0 AWG each branch                          | 2.5 ft each branch (ASSUMED, 4 conductors in rollup) |
| C-06  | SmartShunt load side -> Lynx - bus                                        | 48V | Aggregate return current                                                                                                                    | N/A                                                   | 2/0 AWG                                      | 2.5 ft (ASSUMED)                                     |
| C-06A | Lynx positive tap -> SmartShunt positive sense/power lead                 | 48V | Shunt electronics supply (very low current)                                                                                                 | Factory inline fuse in OEM harness                    | OEM harness lead                             | 2.5 ft (ASSUMED)                                     |
| C-07  | Lynx Slot 1 (F-02) -> MultiPlus DC+                                       | 48V | Inverter branch, fuse-limited                                                                                                               | F-02 125A                                             | 2/0 AWG (manual minimum AWG 1 on short runs) | 2.5 ft (ASSUMED)                                     |
| C-08  | MultiPlus DC- -> Lynx - bus                                               | 48V | Inverter return current                                                                                                                     | F-02 protects paired positive                         | 2/0 AWG                                      | 2.5 ft (ASSUMED)                                     |
| C-09  | MPPT BAT+ -> Lynx Slot 2 (F-03)                                           | 48V | Controller output (45A max)                                                                                                                 | F-03 60A/80V Victron row 188                          | 6 AWG                                        | 2.5 ft (ASSUMED)                                     |
| C-10  | MPPT BAT- -> Lynx - bus                                                   | 48V | Controller return current                                                                                                                   | F-03 protects paired positive                         | 6 AWG                                        | 2.5 ft (ASSUMED)                                     |
| C-11  | Secondary alternator B+ -> APM-48 -> Lynx Slot 3 (F-04)                   | 48V | Alternator branch design current                                                                                                            | F-04 150A                                             | 2/0 AWG                                      | 20 ft (ASSUMED, one-way)                             |
| C-12  | Secondary alternator B- -> Lynx - bus (dedicated return)                  | 48V | Alternator branch return current                                                                                                            | F-04 paired                                           | 2/0 AWG                                      | 20 ft (ASSUMED, one-way)                             |
| C-13  | Lynx / 48V + source -> Orion 48V input protection point                   | 48V | Orion feeder from source, fuse-limited                                                                                                      | Interim F-06 30A; final F-06 20A                      | 6 AWG                                        | 2.5 ft (ASSUMED)                                     |
| C-14  | Orion 48V input protection point -> Orion 48V + input                     | 48V | Orion input, fuse-limited                                                                                                                   | Interim F-06 30A 58V MIDI; final F-06 20A 80V FKS/ATO | 6 AWG planned (8 AWG minimum)                | 2.5 ft (ASSUMED)                                     |

|       |                                                                           |                        |                                                                               |                                                                  | per Orion terminal table)                                   |                                                    |
|-------|---------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------|
| C-15  | Orion 48V - input -> Lynx - bus                                           | 48V                    | Orion input return current                                                    | Orion input positive protection paired                           | 6 AWG                                                       | 2.5 ft (ASSUMED)                                   |
| C-18  | Orion 12V + -> F-07 -> 12V fuse block main + stud                         | 12V                    | Charger output path (30A continuous, 60A fuse)                                | F-07 60A                                                         | 6 AWG planned (8 AWG minimum per Orion table)               | 2.5 ft (ASSUMED)                                   |
| C-19  | Orion 12V - -> 12V fuse block integrated - bus / main - stud              | 12V                    | Charger output return                                                         | F-07 protects paired positive                                    | 6 AWG                                                       | 2.5 ft (ASSUMED)                                   |
| C-19A | 12V buffer battery + -> F-11 -> SW-12V-BATT -> 12V fuse block main + stud | 12V                    | Buffer source path and service isolation path                                 | F-11 100A class                                                  | 4 AWG planned                                               | 2.5 ft (ASSUMED)                                   |
| C-19B | 12V buffer battery - -> 12V fuse block integrated - bus / main - stud     | 12V                    | Buffer battery return path                                                    | N/A (paired with c-19A)                                          | 4 AWG planned                                               | 2.5 ft (ASSUMED)                                   |
| C-20  | 12V panel -> Starlink PSU                                                 | 12V                    | Branch load                                                                   | F-10 10A                                                         | 14 AWG duplex                                               | 8 ft (ASSUMED, near-load branch)                   |
| C-21  | 12V panel -> Fridge                                                       | 12V                    | Branch load                                                                   | F-10 15A                                                         | 14 AWG duplex                                               | 12 ft (ASSUMED, far-load branch)                   |
| C-22  | 12V panel -> LF Bros diesel-heater harness                                | 12V                    | Startup/cooldown branch; keep energized through controller-commanded shutdown | F-10 15A (retain any supplied harness fuse; verify coordination) | 14 AWG duplex only for verified short run; otherwise 12 AWG | 8 ft (ASSUMED; measure before final cable landing) |
| C-23  | 12V panel -> Water pump                                                   | 12V                    | Branch load                                                                   | F-10 10A                                                         | 14 AWG duplex                                               | 8 ft (ASSUMED, near-load branch)                   |
| C-24  | 12V panel -> CO + propane detector                                        | 12V                    | Branch load                                                                   | F-10 3A                                                          | 18/2                                                        | 8 ft (ASSUMED, near-load branch)                   |
| C-25  | 12V panel -> LED lights + dimmer (Hiatus pre-installed)                   | 12V                    | Branch load                                                                   | F-10 5A                                                          | 18/2                                                        | 8 ft (ASSUMED, near-load branch)                   |
| C-26  | 48V system positive/negative -> CERBO-PWR -> Cerbo GX power input         | 48V                    | Cerbo electronics feed (~3W)                                                  | CERBO-PWR 1A-3A inline fuse                                      | 18 AWG red/black duplex acceptable                          | 2.5 ft (ASSUMED, cabinet internal)                 |
| C-27  | PV strings -> F-09 combiner -> MPPT PV input                              | PV string voltage (35) | String current + combiner output current                                      | F-09A/B/C 15A each                                               | 10 AWG PV wire                                              | 12 ft trunk + 3x8 ft string legs (ASSUMED)         |

|      |                                                                                                                    |                                                    |                                                                                |                                                                                    |                                                   |                                        |
|------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------------------|
| C-28 | Shore source/adapters -> portable EMS -> shore cord -> shore inlet -> combined AC DIN enclosure / AC input breaker | 120VAC                                             | Source-limited shore current (adapter-constrained at source)                   | 30A AC input breaker/disconnect baseline with source-current-limit settings policy | 10/3 shore feed to inlet/AC-in area               | 8 ft (ASSUMED)                         |
| C-29 | AC input breaker/disconnect -> MultiPlus AC-in                                                                     | 120VAC                                             | MultiPlus AC input current (30A hardware basis)                                | Upstream 30A AC breaker/disconnect (C-28)                                          | 10 AWG stranded AC conductors                     | 2.5 ft (ASSUMED, cabinet internal)     |
| C-30 | MultiPlus AC-out-1 -> combined AC DIN enclosure / AC-out main breaker                                              | 120VAC                                             | Inverter-backed AC-out feeder current (30A system cap)                         | 30A AC-out main breaker                                                            | 10/3 stranded AC feeder                           | 2.5 ft (ASSUMED, cabinet internal)     |
| C-31 | Branch A -> office/driver GFCI receptacle                                                                          | 120VAC                                             | Branch load (office monitor/chargers/general outlet use)                       | 20A branch breaker + GFCI receptacle                                               | 12/3 stranded AC branch cable                     | 15 ft (ASSUMED, branch leg default)    |
| C-32 | Branch B -> galley/passenger GFCI receptacle                                                                       | 120VAC                                             | Branch load (galley/general high-draw outlet; induction/Ninja SP151 sequenced) | 20A branch breaker + GFCI receptacle                                               | 12/3 stranded AC branch cable                     | 15 ft (ASSUMED, branch leg default)    |
| C-33 | MultiPlus AC-out-2 (reserve-only) -> capped route for future shore-only branch                                     | 120VAC                                             | N/A in Phase 1 (route reserved only)                                           | N/A in Phase 1 (no energized branch hardware)                                      | 12 AWG stranded AC conductors (reserve path only) | 15 ft (ASSUMED, reserve route default) |
| C-34 | 12V panel -> USB PD station branch (office zone)                                                                   | 12V                                                | High-demand office charging branch (100W + 65W class station budget)           | F-10 branch fuse (20A)                                                             | 12 AWG duplex baseline                            | 5 ft (ASSUMED, short-run requirement)  |
| C-35 | 12V panel -> USB PD station branch (galley zone)                                                                   | 12V                                                | Galley charging branch (65W class USB-C plus USB-A/C loads)                    | F-10 branch fuse (15A)                                                             | 14 AWG duplex baseline                            | 8 ft (ASSUMED, near-load branch)       |
| C-36 | 12V panel -> Maxxair fan (Hiatus pre-installed)                                                                    | 12V                                                | Roof ventilation branch                                                        | F-10 branch fuse (10A)                                                             | 14 AWG duplex baseline                            | 8 ft (ASSUMED, near-load branch)       |
| C-37 | 12V panel -> DC ambient/cabinet LED strips (planned Govee)                                                         | 12V                                                | Branch load                                                                    | F-10 branch fuse (5A)                                                              | 18/2 baseline                                     | 8 ft (ASSUMED, near-load branch)       |
| C-38 | WS500 regulator power source -> WS500 regulator input                                                              | 48V-referenced unless harness docs prove otherwise | Regulator electronics feed                                                     | F-12 (10A/15A)                                                                     | Harness lead                                      | 8 ft (ASSUMED)                         |
| C-39 | WS500 battery positive-sense lead -> WS500                                                                         | 48V sense lead                                     | Regulator voltage sense                                                        | F-13 (3A)                                                                          | Harness lead                                      | 8 ft (ASSUMED)                         |

|         |                                                                                                              |                                               |                                                              |                                                              |                                                                              |                                                                   |
|---------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------|
| C-40    | WS500 current-sense high/low pair to selected shunt/current-sense point                                      | low-current sense                             | Regulator current feedback                                   | No fuse per current Wakespeed manual; twist pair if extended | Harness lead                                                                 | 8 ft (ASSUMED)                                                    |
| C-41    | Ford Upfitter #3 output -> F-15 -> WS500 brown ignition/enable wire                                          | 12V control lead                              | Manual alternator-enable signal only                         | F-15 (3A)                                                    | 16 AWG TXL/GXL                                                               | 6 ft (ASSUMED)                                                    |
| C-42    | 12V panel -> Kicker 46KMC2 media center/source unit                                                          | 12V                                           | Source/head-unit branch (15A max; KMC2 manual shows 15A ATM) | F-10/AUDIO-HU 15A branch plus KMC2 harness fuse              | 12 AWG duplex if kept near 5 ft; use 10 AWG if longer                        | 5 ft (ASSUMED, driver-side DC shelf)                              |
| C-43    | 12V source/main + stud -> AUDIO-SUB source fuse -> Kicker 49PTRTP10 powered sub +                            | 12V                                           | Powered sub branch; PTRTP10 manual external fuse value       | AUDIO-SUB 40A                                                | 4 AWG tinned/OFC                                                             | 8 ft (ASSUMED; shorten by placing sub near 12V junction)          |
| C-44    | Kicker 49PTRTP10 powered sub - -> 12V negative bus/main - stud                                               | 12V                                           | Powered sub return                                           | AUDIO-SUB protects paired positive                           | 4 AWG tinned/OFC                                                             | 8 ft (ASSUMED, paired with positive)                              |
| C-45    | KMC2 remote turn-on output -> PTRTP10 remote input                                                           | 12V low-current control                       | Remote amp/sub enable                                        | KMC2/source branch protected                                 | 18 AWG                                                                       | 8 ft (ASSUMED)                                                    |
| C-46    | KMC2 RCA line-out -> PTRTP10 RCA input                                                                       | low-level audio signal                        | Subwoofer signal                                             | N/A                                                          | shielded 2-channel RCA                                                       | measured route (~4m planning cable)                               |
| C-47L/R | KMC2 front speaker outputs -> left/right Kicker CSC67 speakers                                               | speaker-level audio                           | One 4 ohm speaker per KMC2 channel                           | KMC2 internal protection / AUDIO-HU source branch            | 16 AWG marine speaker wire                                                   | 10-12 ft each (ASSUMED)                                           |
| C-48    | KUS SSS/SSL sender black signal + pink return -> matching signal/negative pair in Cerbo GX MK2 Tank 1 column | Cerbo low-current resistive-sender excitation | Fresh-water level, US 240-30 ohm                             | No external fuse or power feed; do not share chassis return  | 18-22 AWG tinned duplex, ferrules at Cerbo, sealed pigtail splices at sender | Measure after tank/Cerbo endpoints are installed (ASSUMED <10 ft) |

## Wiring Validation Worksheet (Estimate Pass, 2026-02-18)

Calculation basis for drop screening:

- $V_{drop} = I * (2 * L_{one\_way} * R_{per\_ft})$
- Resistance basis used in this pass ( ohm/ft ): 2/0=0.0000779 , 6 AWG=0.0003951 , 4 AWG=0.0002485 , 12 AWG=0.001588 , 14 AWG=0.002525 , 18/2=0.006385 , 10 AWG=0.000999 .
- Design targets: <=2% on major 48V trunks, <=3% on planned 12V /AC branches.

| 4. C-05 is a rollup row that includes both branch and trunk return paths; voltage-drop screen shown is the conservative interim worst-case ( 155A ) within that rollup. |                    |                                       |                  |                                                                                                      |         |                                                                                                           |                                     |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------------------------------------|------------------|------------------------------------------------------------------------------------------------------|---------|-----------------------------------------------------------------------------------------------------------|-------------------------------------|--------|
| Circuit ID                                                                                                                                                              | From               | To                                    | Fuse             | Current basis                                                                                        | Gauge   | Estimated one-way length                                                                                  | Voltage drop %                      | BC     |
| C-01                                                                                                                                                                    | Battery A +        | F-01A                                 | F-01A 200A       | 77.5A interim design branch share                                                                    | 2/0 AWG | 2.5 ft planning placeholder; field layout may use unequal positive lengths if total loop path is balanced | 0.059% @ 51.2V planning placeholder | Ro red |
| C-02                                                                                                                                                                    | Battery B +        | F-01B                                 | F-01B 200A       | 77.5A interim design branch share                                                                    | 2/0 AWG | 2.5 ft planning placeholder; field layout may use unequal positive lengths if total loop path is balanced | 0.059% @ 51.2V planning placeholder | Ro red |
| C-02C                                                                                                                                                                   | Battery C +        | F-01C                                 | F-01C 200A       | 77.5A interim design branch share                                                                    | 2/0 AWG | 2.5 ft planning placeholder; field layout may use unequal positive lengths if total loop path is balanced | 0.059% @ 51.2V planning placeholder | Ro red |
| C-03                                                                                                                                                                    | Class T load studs | 48V + bus / disconnect input          | F-01A/B/C        | 77.5A interim per branch                                                                             | 2/0 AWG | 2.5 ft each (x4 conductors)                                                                               | 0.059% @ 51.2V                      | Ro red |
| C-04                                                                                                                                                                    | Disconnect output  | Lynx + bus                            | Upstream Class T | 155A interim / 145A final aggregate                                                                  | 2/0 AWG | 2.5 ft                                                                                                    | 0.12% @ 51.2V interim               | Ro red |
| C-05                                                                                                                                                                    | Battery - branches | SmartShunt battery side via 48V - bus | N/A              | 77.5A per battery-negative branch; row rollup also includes one 155A interim trunk (NEGBUS_TO_SHUNT) | 2/0 AWG | 2.5 ft each (x4 conductors)                                                                               | 0.12% @ 51.2V (worst-case rollup)   | Ro bla |

|       |                              |                                           |                                                    |                                            |               |        |                            |        |
|-------|------------------------------|-------------------------------------------|----------------------------------------------------|--------------------------------------------|---------------|--------|----------------------------|--------|
| C-06  | SmartShunt load side         | Lynx - bus                                | N/A                                                | 155A interim / 145A final aggregate return | 2/0 AWG       | 2.5 ft | 0.12% @ 51.2V interim      | Ro bla |
| C-06A | Lynx positive tap            | SmartShunt sense/power lead               | OEM inline fuse                                    | OEM harness current                        | OEM harness   | 2.5 ft | N/A (low-current OEM lead) | Ro ha  |
| C-07  | Lynx Slot 1 DC+              | MultiPlus DC+                             | F-02 125A                                          | 125A                                       | 2/0 AWG       | 2.5 ft | 0.10% @ 51.2V              | Ro red |
| C-08  | MultiPlus DC-                | Lynx - bus                                | F-02 paired                                        | 125A                                       | 2/0 AWG       | 2.5 ft | 0.10% @ 51.2V              | Ro bla |
| C-09  | MPPT BAT+                    | Lynx Slot 2                               | F-03 60A (80V Victron row 188)                     | 45A                                        | 6 AWG         | 2.5 ft | 0.17% @ 51.2V              | Ro red |
| C-10  | MPPT BAT-                    | Lynx - bus                                | F-03 paired                                        | 45A                                        | 6 AWG         | 2.5 ft | 0.17% @ 51.2V              | Ro bla |
| C-11  | Secondary alternator B+      | Lynx Slot 3 via APM-48                    | F-04 150A                                          | 150A design                                | 2/0 AWG       | 20 ft  | 0.80% @ 58.4V              | Ro red |
| C-12  | Secondary alternator B-      | Lynx - bus (dedicated return)             | F-04 paired                                        | 150A design                                | 2/0 AWG       | 20 ft  | 0.80% @ 58.4V              | Ro bla |
| C-13  | Lynx 48V + bus tap           | F-06 source side                          | Interim F-06 30A 58V MIDI; final 20A 80VDC FKS/ATO | 30A interim / 20A final                    | 6 AWG         | 2.5 ft | 0.08% @ 51.2V              | Ro red |
| C-14  | Orion input protection point | Orion 48V +                               | Interim F-06 30A 58V MIDI; final 20A 80VDC FKS/ATO | 30A interim / 20A final                    | 6 AWG         | 2.5 ft | 0.08% @ 51.2V              | Ro red |
| C-15  | Orion 48V -                  | Lynx - bus                                | Orion input positive protection paired             | 30A interim / 20A final fuse basis         | 6 AWG         | 2.5 ft | 0.08% @ 51.2V              | Ro bla |
| C-18  | Orion 12V +                  | Fuse block main + stud                    | F-07 60A/80V Victron row 188                       | 30A                                        | 6 AWG         | 2.5 ft | 0.49% @ 12V                | Ro red |
| C-19  | Orion 12V -                  | Fuse block integrated - bus / main - stud | F-07 paired                                        | 30A                                        | 6 AWG         | 2.5 ft | 0.49% @ 12V                | Ro bla |
| C-19A | Buffer battery +             | Fuse block main + stud (via F-11/SW)      | F-11 100A                                          | 50A design                                 | 4 AWG         | 2.5 ft | 0.52% @ 12V                | Ro red |
| C-19B | Buffer battery -             | Fuse block integrated - bus / main - stud | N/A                                                | 50A design                                 | 4 AWG         | 2.5 ft | 0.52% @ 12V                | Ro bla |
| C-20  | 12V fuse panel               | Starlink PSU                              | F-10 10A                                           | 8A                                         | 14 AWG duplex | 8 ft   | 2.69% @ 12V                | Ro AWG |

|      |                             |                                            |                                             |                    |                                                    |                                  |                                    |                               |
|------|-----------------------------|--------------------------------------------|---------------------------------------------|--------------------|----------------------------------------------------|----------------------------------|------------------------------------|-------------------------------|
| C-21 | 12V fuse panel              | Fridge                                     | F-10 15A                                    | 7A                 | 14 AWG duplex                                      | 12 ft                            | 3.54% @ 12V                        | Ro<br>AWG                     |
| C-22 | 12V fuse panel              | LF Bros diesel-heater harness              | F-10 15A + supplied harness fuse if present | 10A startup screen | 14 AWG duplex short-run candidate; 12 AWG fallback | 8 ft assumed                     | 3.37% @ 12V with 14 AWG            | Ro<br>AWG<br>pla<br>AW<br>rec |
| C-23 | 12V fuse panel              | Water pump                                 | F-10 10A                                    | 7A                 | 14 AWG duplex                                      | 8 ft                             | 2.36% @ 12V                        | Ro<br>AWG                     |
| C-24 | 12V fuse panel              | CO+propane detector                        | F-10 3A                                     | 0.2A               | 18/2                                               | 8 ft                             | 0.17% @ 12V                        | Ro                            |
| C-25 | 12V fuse panel              | LED lights + dimmer (Hiatus pre-installed) | F-10 5A                                     | 5A                 | 18/2                                               | 8 ft                             | 4.26% @ 12V                        | Ro                            |
| C-26 | 48V system feed             | Cerbo GX                                   | CERB0-PWR 1A-3A                             | ~3W                | 18 AWG duplex acceptable                           | 2.5 ft                           | N/A (low-current electronics feed) | Low<br>ins<br>rov             |
| C-27 | PV strings/combiner         | MPPT PV input                              | F-09A/B/C 15A                               | 30A trunk screen   | 10 AWG PV                                          | 12 ft trunk + 3x8 ft string legs | 0.72% @ 100V trunk screen          | Ro<br>AW<br>eq                |
| C-28 | Shore inlet path            | AC input breaker/disconnect                | Source-limited AC OCP                       | 30A hardware basis | 10/3                                               | 8 ft                             | 0.40% @ 120VAC                     | Ro<br>sho<br>in/              |
| C-29 | AC input breaker/disconnect | MultiPlus AC-in                            | Upstream AC OCP                             | 30A hardware basis | 10 AWG AC                                          | 2.5 ft                           | 0.12% @ 120VAC                     | Ro<br>sho<br>in/              |
| C-30 | MultiPlus AC-out-1          | AC-out main breaker                        | 30A AC-out main                             | 30A hardware basis | 10/3                                               | 2.5 ft                           | 0.12% @ 120VAC                     | Ro<br>sho<br>in/              |
| C-31 | Branch A                    | Office/driver receptacle chain             | 20A branch OCP                              | 20A                | 12 AWG AC                                          | 15 ft                            | 0.79% @ 120VAC                     | Ro<br>AWG                     |
| C-32 | Branch B                    | Galley/passenger receptacle chain          | 20A branch OCP                              | 20A                | 12 AWG AC                                          | 15 ft                            | 0.79% @ 120VAC                     | Ro<br>AWG                     |
| C-33 | MultiPlus AC-out-2          | Reserve-only capped route                  | N/A in Phase 1                              | N/A                | 12 AWG AC (reserve path only)                      | 15 ft                            | 0.60% @ 120VAC (future-use screen) | N/<br>pro<br>Ph               |
| C-34 | 12V fuse panel              | Office USB PD station                      | F-10 20A                                    | 20A design cap     | 12 AWG duplex                                      | 5 ft                             | 2.65% @ 12V                        | Ro<br>AWG<br>USE              |

|           |                                     |                                               |                                                  |                                                                        |                                  |                |                       |                  |
|-----------|-------------------------------------|-----------------------------------------------|--------------------------------------------------|------------------------------------------------------------------------|----------------------------------|----------------|-----------------------|------------------|
| C-35      | 12V fuse panel                      | Galley USB PD station                         | F-10 15A                                         | 8A expected                                                            | 14 AWG duplex                    | 8 ft           | 2.69% @ 12V           | Ro<br>AWG<br>USB |
| C-36      | 12V fuse panel                      | Maxxair fan (Hiatus pre-installed)            | F-10 10A                                         | 4A expected                                                            | 14 AWG duplex                    | 8 ft           | 1.35% @ 12V           | Ro<br>AWG        |
| C-37      | 12V fuse panel                      | DC ambient/cabinet LED strips (planned Govee) | F-10 5A                                          | 5A design cap                                                          | 18/2                             | 8 ft           | 4.26% @ 12V           | Ro               |
| C-38      | WS500 regulator power source        | WS500 regulator input                         | F-12 10A/15A                                     | low-current electronics feed; likely 48V-referenced in this build      | Harness lead                     | 8 ft           | N/A (harness-limited) | Ro<br>kit        |
| C-39      | WS500 positive voltage-sense source | WS500 voltage-sense input                     | F-13 3A                                          | low-current 48V sense lead; fuse/holder voltage class must be verified | Harness lead                     | 8 ft           | N/A (harness-limited) | Ro<br>kit        |
| C-40      | WS500 current-sense high/low pair   | WS500 current-sense input                     | No fuse per current manual                       | low-current sense; twist pair if extended                              | Harness lead                     | 8 ft           | N/A (harness-limited) | Ha               |
| C-41      | Ford Upfitter #3 output             | WS500 brown ignition/enable input via F-15    | F-15 3A                                          | manual low-current control only                                        | 16 AWG TXL/GXL                   | 6 ft           | N/A (control circuit) | Ro<br>(up<br>con |
| C-42      | 12V panel                           | Kicker 46KMC2 media center                    | AUDIO0-HU 15A branch + KMC2 15A ATM harness fuse | 15A max fuse basis                                                     | 12 AWG duplex                    | 5 ft           | 1.99% @ 12V           | Ro<br>au         |
| C-43/C-44 | 12V source/main studs               | Kicker 49PTRTP10 powered sub                  | AUDIO0-SUB 40A external source fuse              | 40A manual fuse basis                                                  | 4 AWG positive + matching return | 8 ft           | 1.33% @ 12V           | Ro<br>au<br>kit  |
| C-45      | KMC2 remote output                  | PTRTP10 remote input                          | KMC2/source branch protected                     | low-current remote                                                     | 18 AWG                           | 8 ft           | N/A                   | Ro<br>au<br>sig  |
| C-46      | KMC2 RCA line-out                   | PTRTP10 RCA input                             | N/A                                              | low-level signal                                                       | shielded RCA                     | measured route | N/A                   | Ro<br>au<br>sig  |
| C-47L/R   | KMC2 front speaker outputs          | Left/right csc67 speakers                     | KMC2/source branch protected                     | one 4 ohm speaker per channel                                          | 16 AWG marine speaker wire       | 10-12 ft each  | N/A                   | Ro<br>au<br>spe  |

# Wire Rollup (No-Padding Purchase Baseline)

| Gauge / cable family                       | Estimated total                                                                    | Source circuits                                                                                                                               | BOM row                |
|--------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 2/0 AWG red                                | 42.5 ft                                                                            | C-01, C-02, C-02C, C-03, C-04, C-07, C-11                                                                                                     | 28                     |
| 2/0 AWG black                              | 35.0 ft                                                                            | C-05, C-06, C-08, C-12                                                                                                                        | 28                     |
| 6 AWG red                                  | 10.0 ft                                                                            | C-09, C-13, C-14, C-18                                                                                                                        | 29                     |
| 6 AWG black                                | 7.5 ft                                                                             | C-10, C-15, C-19                                                                                                                              | 29                     |
| 4 AWG red                                  | 10.5 ft (2.5 ft existing + 8 ft audio sub planning branch)                         | C-19A, C-43                                                                                                                                   | 30, 192                |
| 4 AWG black                                | 10.5 ft (2.5 ft existing + 8 ft audio sub return)                                  | C-19B, C-44                                                                                                                                   | 30, 192                |
| 10 AWG pair-equivalent (PV)                | placeholder only                                                                   | C-27 final module/string topology deferred until solar workstream; do not buy combiner/fuse count or cut roof entries from the old 3S3P model | 31                     |
| 14 AWG duplex                              | 52 ft                                                                              | C-20, C-21, C-22, C-23, C-35, C-36                                                                                                            | 32                     |
| 18/2                                       | 24.0 ft plus separate Cerbo low-current duplex                                     | C-24, C-25, C-37; Cerbo C-26 uses 48V fused low-current duplex                                                                                | 33 / low-current stock |
| 12 AWG AC branch cable                     | 30 ft (C-33 excluded in Phase 1)                                                   | C-31, C-32                                                                                                                                    | 113                    |
| 10/3 shore + AC-in/out feed                | 13 ft                                                                              | C-28, C-29, C-30                                                                                                                              | 114                    |
| USB/audio branch mix (12 AWG + 14 AWG)     | USB: 5 ft (12 AWG) + 8 ft (14 AWG); audio source: 5 ft 12 AWG short-run assumption | C-34, C-35, C-42                                                                                                                              | 116, 193               |
| Camper audio signal/speaker/control wiring | RCA measured route, 18 AWG remote, and 16 AWG marine speaker wire runs             | C-45, C-46, C-47L/R                                                                                                                           | 193                    |
| WS500 harness/sense leads                  | Included in selected kit/harness set                                               | C-38, C-39, C-40                                                                                                                              | 168, 171               |
| 16 AWG TXL/GXL control wire                | 6 ft                                                                               | C-41                                                                                                                                          | 176                    |

Notes:

- 1. This table is a base estimate only; it intentionally excludes order padding and termination waste.
- 2. Apply personal order overage at checkout based on actual spool cut increments and routing confidence.
- 3. Parallel bank balancing is locked by similar total loop resistance per battery. Positive-only leads may differ if the paired negative path offsets the difference; verify with final measured lengths and, if needed, clamp-current checks.

## 3x Battery Bank Bench-Build Cut List (2/0 AWG)

Purpose: make the bench build orderable without needing final camper run lengths. Treat lengths below as *bench module* lengths only; final install harnesses should be re-cut after layout freeze.

Assumptions:

- 1. Battery terminals are M8 (verify your battery stud size before ordering lugs).
- 2. Class T fuse blocks and battery-side busbars use 3/8" studs (treat as M10 lugs unless your specific hardware differs).
- 3. Lynx Distributor is the M10 model (main connections M10 ; internal/fuse studs may still be M8 depending on the position).

| Cable ID | Qty | From -> To | Color | Gauge | Est. one-way length | Lug A | Lug B |
|----------|-----|------------|-------|-------|---------------------|-------|-------|
|----------|-----|------------|-------|-------|---------------------|-------|-------|

|                      |   |                                         |       |     |                       |     |     |
|----------------------|---|-----------------------------------------|-------|-----|-----------------------|-----|-----|
| BATT+_A/B/C          | 3 | Battery + -> Class T block line side    | red   | 2/0 | 2.5 ft each (ASSUMED) | M8  | M10 |
| FUSE_TO_POSBUS_A/B/C | 3 | Class T block load side -> 48V + busbar | red   | 2/0 | 2.5 ft each (ASSUMED) | M10 | M10 |
| POSBUS_TO_DISC       | 1 | 48V + busbar -> disconnect input        | red   | 2/0 | 2.5 ft (ASSUMED)      | M10 | M10 |
| DISC_TO_LYNX+        | 1 | disconnect output -> Lynx + input       | red   | 2/0 | 2.5 ft (ASSUMED)      | M10 | M10 |
| BATT-_A/B/C          | 3 | Battery - -> 48V - busbar               | black | 2/0 | 2.5 ft each (ASSUMED) | M8  | M10 |
| NEGBUS_TO_SHUNT      | 1 | 48V - busbar -> SmartShunt battery side | black | 2/0 | 2.5 ft (ASSUMED)      | M10 | M10 |
| SHUNT_TO_LYNX-       | 1 | SmartShunt load side -> Lynx - input    | black | 2/0 | 2.5 ft (ASSUMED)      | M10 | M10 |
| LYNX_SLOT1_TO_MULTI+ | 1 | Lynx Slot 1 DC+ -> MultiPlus DC+        | red   | 2/0 | 2.5 ft (ASSUMED)      | M8  | M8  |
| LYNX_TO_MULTI-       | 1 | Lynx - -> MultiPlus DC-                 | black | 2/0 | 2.5 ft (ASSUMED)      | M8  | M8  |

Locked balancing rule for the 3x parallel bank:

1. Keep each battery path's total positive + negative loop resistance similar.
2. Equal positive-only leads are not required when the negative lead lengths intentionally offset the positive lead differences.
3. Keep the same cable family, lug geometry, fuse/holder family, and termination quality across all three battery paths.
4. Verify sharing with clamp-current checks under charge/load if final measured path lengths differ materially.

Torque reference (verify against your exact manuals/hardware):

- MultiPlus-II DC terminals: 12 Nm ( M8 nut) per Victron installation guidance.
- SmartShunt shunt bolts: verify torque for the installed 300A model per Victron installation guidance.
- Lynx Distributor M10 model: M10 nuts 33 Nm (older serials may be lower), and M8 nuts 14 Nm per Victron Lynx installation guidance.

## Additional Components Included In Topology Scope

- 48V disconnect ( 275A )
- Pre-charge resistor (commissioning/soft-charge aid before connecting large DC loads)
- Battery-side 48V + combine busbar (after Class T fuses)
- Battery-side 48V - combine busbar (battery-only, before SmartShunt)
- 12V fuse block main + stud used as source-combine point (Orion + buffer battery feed)
- 12V fuse block integrated negative bus/main - used as shared return point
- 12V buffer battery main fuse ( F-11 ) and manual disconnect switch ( SW-12V-BATT )
- Shore AC inlet + cord/adapter interface hardware
- Portable EMS in source-side shore path before camper inlet
- Single combined 6-way AC DIN enclosure with isolated AC-in and AC-out neutral paths plus common equipment grounding/PE handling
- AC-out 30A main breaker plus 20A / 20A branch breaker and GFCI receptacle hardware
- Receptacle boxes + 120V outlets (current purchased baseline 2 GFCI receptacles; row 112 remains planned/confirm-on-hand for boxes/covers)
- AC-out-2 reserve-only capped route (no energized Phase 1 branch hardware)
- USB PD station branch hardware ( 2 stations: office + galley)
- Camper audio hardware: Kicker 46KMC2 source branch, Kicker 49PTRTP10 powered sub branch, AUDIO-HU 15A source protection, AUDIO-SUB 40A source fuse, RCA/speaker/remote wiring
- Battery temperature sensor wiring to inverter/monitoring path
- SmartShunt fused positive sense/power lead (factory harness)
- Cerbo GX CERB0-PWR small inline fused 48V power feed
- Ford Upfitter #3 control lead and local F-15 inline fuse for the WS500 brown ignition/enable wire

## Assumptions (Explicit)

1. Cable sizing assumes fine-strand copper conductors (OFC welding-cable baseline for high-current DC paths), enclosed vehicle routing, and the estimated one-way lengths listed in this document.
2. Voltage-drop design intent used here: <=2% on major 48V power runs and <=3% on 12V branch circuits.
3. F-09 PV string fuse value ( 15A ) remains provisional until final module datasheet max-series-fuse rating is confirmed.
4. Cerbo GX feed is now a small inline fused 48V feed ( CERB0-PWR ) from the system/load side of the main disconnect during bench commissioning, so the Cerbo powers down with the house system.

5. Orion branch uses one source-side F-06 from a Lynx 48V+ bus tap during build: existing 30A 58V MIDI now, planned 20A 80V FKS/ATO cleanup later. Lynx Slot 4 ( F-05 ) remains open/blank; do not add a duplicate Orion MEGA fuse unless the topology is deliberately reopened.
6. No low-voltage-disconnect (LVD) automation is included in Phase 1; protection is source fusing plus manual SW-12V-BATT isolation.
7. Alternator architecture lock is dedicated 48V secondary alternator path ( Mechman + WS500 + APM-48 ) with F-04 150A ; obsolete pre-Mechman engine-bay fuse paths are removed from active layout.
8. F-01A/B/C are provisionally set to 200A pending final 51.2V battery datasheet/manual confirmation; if validated limits are lower, shift to 175A .
9. 2/0 cable quantity planning baseline in this pass is 77.5 ft total no-padding ( 42.5 ft red + 35.0 ft black); user-applied order padding is intentionally deferred to checkout.
10. Manual alternator shutdown baseline is Ford Upfitter #3 feeding the WS500 brown ignition/enable wire through F-15 ; WS500 white Feature-In remains a future-only reserve for automatic interlock work.
11. Camper audio is a 12V accessory load. KMC2 source branch is 15A ; PTRTP10 sub branch is 40A with 4 AWG power and return. Audio peaks can exceed Orion continuous output and rely on the 12V buffer battery, so first loud testing should watch 12V voltage/SOC.

## Completion Status

- DC/PV topology is complete for current BOM scope and load model scope.
- AC hierarchy is complete at architecture level, including transfer behavior, branch strategy, and protection chain.
- Full-circuit estimate pass is now documented with run lengths, voltage-drop screening, and purchase rollups.
- Camper audio branch added as a DC-first 12V accessory package: Kicker 46KMC2 source branch plus Kicker 49PTRTP10 powered sub branch; detailed product/routing notes live in docs/implementation/CAMPER\_audio\_system.md .
- Remaining work is final measured-length replacement and SKU-level closeout before final cut-to-length harness production.